

BeeEcoTox: A Large Language Model-Empowered Graph-Learning Framework for Predicting Ecotoxicity of Agrochemicals to Bees

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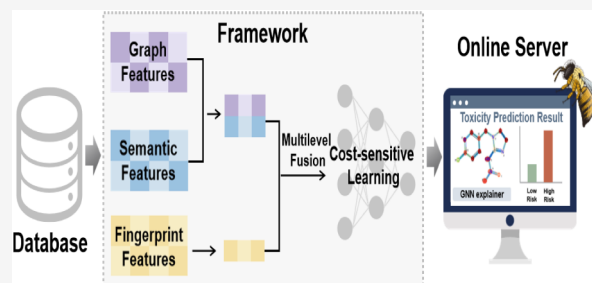
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ABSTRACT: Agrochemical exposure can threaten bees with substantial ecological risks. The rapid and accurate prediction of agrochemical ecotoxicity to bees is thus urgently needed; however, existing models are constrained by single-type structural inputs, resulting in limited accuracy and generalizability. This study presents BeeEcoTox, a multimodal graph-learning framework for predicting agrochemical ecotoxicity to bees. The model fuses ChemFM-derived semantic features with molecular graphs through a graph isomorphism network with internal batch normalization, combined with structural features from 1,139 agrochemicals. Inherent class imbalance in the curated data set is addressed through a cost-sensitive learning approach to ensure that the model prioritizes high recall in detecting ecotoxic agrochemicals without compromising overall performance. BeeEcoTox achieves state-of-the-art performance, with area under the curve and recall values of 0.91 and 0.90, respectively, following rigorous benchmarking against a suite of baseline models. Model explainability is enhanced through the use of GNNExplainer, a model-agnostic approach for identifying key toxicophores. BeeEcoTox is deployed as a publicly available web-based platform (<https://www.ai4environ.cn/BeeEcoTox>) to support advanced ecological risk assessments of agrochemicals and the development of new approach methodologies.

KEYWORDS: ecotoxicity, agrochemicals, deep learning, large language model, multimodal fusion, risk assessment



1. INTRODUCTION

Extensive agrochemical application poses substantial ecological risks to nontarget animals, plants, and microorganisms.^{1–5} Acute and chronic exposures to agrochemicals can adversely affect organisms across trophic levels and organizational scales, from individual development and population dynamics to ecosystem functioning.^{6–9} These widespread impacts highlight the need for systematic assessments of agrochemical ecotoxicity. With rapid expansion of the number and volume of applied agrochemicals and increasingly rigorous regulatory requirements, predictive models have the potential to become an indispensable tool for high-throughput ecotoxicity evaluations.

The evolution of predictive approaches for various types of ecotoxicity has progressed from early quantitative structure–activity relationship models and molecular simulations to contemporary artificial intelligence-based strategies. These methods have facilitated ecotoxicity screening across diverse biological taxa,^{10–16} with particular emphasis on bees owing to their ecological and agricultural significance. Existing models for bee ecotoxicity prediction predominantly leverage machine

learning and deep learning techniques, including ensemble methods, multidescrptor frameworks, and graph neural networks (GNNs).^{17–24} A key challenge in model construction is the generation of adequate discriminative features or modalities that reflect the complex structural and semantic information governing ecotoxicity. Current approaches remain constrained by their considerable reliance on predefined engineered features or graph topologies that cannot sufficiently incorporate semantically rich molecular representations.²⁵

The success of domain-specific chemical language models (CLMs) trained on extensive chemical corpora,^{26–30} such as ChemFM, has opened new opportunities for providing deep, context-aware molecular representations that capture rich chemical semantics from Simplified Molecular Input Line

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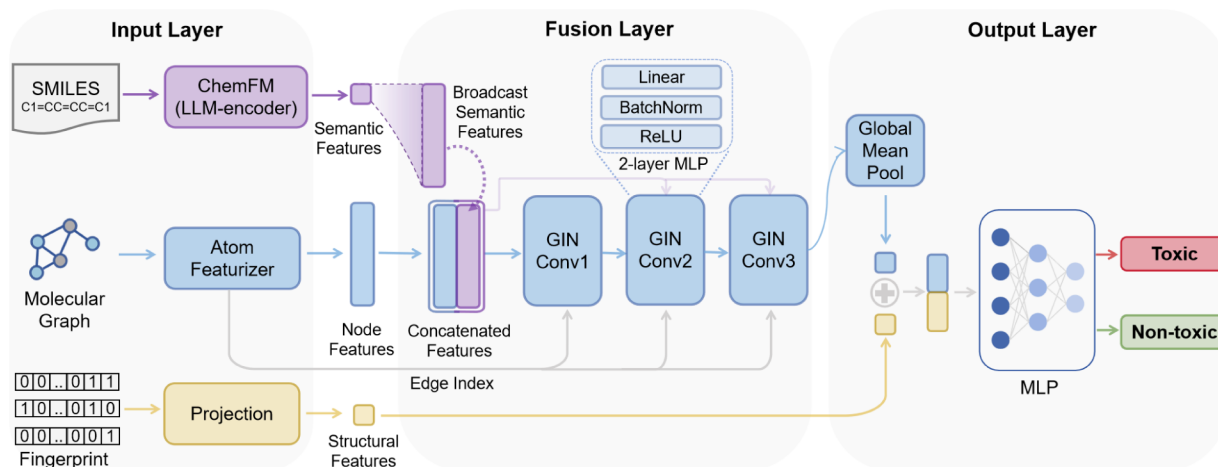


Figure 1. Schematic architecture of BeeEcoTox. The multimodal model integrates molecular graphs, ChemFM-derived semantic embeddings, and MACCS structural fingerprints. In the fusion layer, global semantic features are broadcast and concatenated with atomic features, then processed through a 3-layer graph isomorphism network (GIN) incorporating internal batch normalization. Each layer utilizes a multilayer perceptron with successive linear transformations, batch normalization, and ReLU activation. The resulting graph representation is aggregated via global mean pooling, concatenated with projected fingerprint features, and processed by a fully connected MLP for binary toxicity classification.

Entry System (SMILES) strings.^{31–33} These semantic representations provide powerful complementary information that enhances structural features, bridging the gap between symbolic chemical notation and graph-based molecular topology. The integration of LLM-derived semantic features with structural graph representations and functional fingerprints can help to establish multimodal models, with each modality contributing complementary molecular information. This unification of structural, semantic, and functional perspectives, and their multimodal integration, opens new avenues for developing robust and explainable models that are essential for predicting chemical ecotoxicity.

This study presents a semantically enriched, multimodal graph-learning framework for predicting agrochemical ecotoxicity to bees, named BeeEcoTox. This framework executes ecotoxicity prediction through the hierarchical fusion of molecular graph features with semantic embeddings, which are generated by encoding SMILES strings using ChemFM via a graph isomorphism network (GIN) with internal batch normalization (Figure 1). A cost-sensitive learning approach was integrated during model training to mitigate the detrimental effects of the inherent class imbalance within the ecotoxicity data set curated from multiple public sources. The framework was rigorously benchmarked against several baseline models to evaluate model performance. Model interpretability was enhanced by employing GNNExplainer to identify critical substructures and atomic contributions to ecotoxicity. To facilitate its application, BeeEcoTox was deployed on a publicly available cloud server. The proposed framework provides a practical and highly efficient alternative approach for advanced ecological risk assessments of agrochemicals.

2. MATERIALS AND METHODS

2.1. Data Collection

Acute toxicity data for agrochemicals were primarily retrieved from public sources, including ECOTOX, the Pesticide Properties Database (PPDB),³⁴ and the PAN Pesticide Database (<https://www.pesticideinfo.org>). For data retrieved from ECOTOX, records were strictly filtered to include only laboratory-based acute toxicity studies. While the majority of records are associated with *Apis mellifera*, toxicity data from other bee species, although limited, were

also included to align with the study's scope of assessing general bee ecotoxicity. The recently published ApisTox³⁵ data set was utilized as a supplementary resource to enrich the chemical space. To ensure data integrity, all molecular structures were standardized using RDKit to generate canonical SMILES strings.

The median lethal dose (LD₅₀) was selected as the toxicity end point, consistent with regulatory guidelines from the U.S. Environmental Protection Agency, European Union, and United Kingdom. A conservative approach based on the precautionary principle was adopted to address the inherent variability in the curated data set, with the lowest value selected to represent the maximum potential hazard for agrochemicals when multiple LD₅₀ values were reported. To avoid conversion errors, only records reported in $\mu\text{g}/\text{bee}$ or $\mu\text{g}/\text{org}$ were utilized for comparison against the 11 $\mu\text{g}/\text{bee}$ threshold. Exposure durations typically did not exceed 72 h, in accordance with standard acute toxicity testing protocols. A unified ecotoxicity classification was applied regardless of bee species or exposure route: an agrochemical was classified as ecotoxic if any available record indicated $\text{LD}_{50} \leq 11 \mu\text{g}/\text{bee}$ and as nonecotoxic to bees if all available data consistently reported $\text{LD}_{50} > 11 \mu\text{g}/\text{bee}$. This workflow yielded a final data set of 1,139 unique agrochemicals (Table S1), which completely encompasses the ApisTox benchmark and provides an additional 104 unique compounds (Figure S1).

2.2. Data Set Processing

A stratification strategy was implemented using fundamental physicochemical descriptors to partition the data into subsets that were balanced in ecotoxicity labels and chemical space. This approach mitigated the limitations of random splits, which can distort class ratios, and of label-only stratification, which may perpetuate biases in underlying molecular property distributions. Three molecular descriptors were selected owing to their established influence on environmental fate, bioavailability, and ecotoxic action, leading to their common use in environmental ecotoxicology and predictive modeling.^{36,37} These descriptors included molecular weight, LogP, and heavy atom count.

The stratification strategy followed a rigorous two-stage procedure. First, continuous values of the selected descriptors were discretized into binary categories based on their median values across the entire data set. These discretized property bins were then combined with the binary ecotoxicity labels by concatenating their values into a single composite string variable. This string (e.g., "1_0_1_1") effectively served as a unique stratum identifier for each agrochemical. Second, the data set was partitioned through hierarchical sampling into training, validation, and test sets in an 8:1:1 ratio. Specifically, the

training set was first separated to contain 80% of the total data, and the remaining holdout set was subsequently divided into equal validation and test subsets. This procedure ensured consistent class ratios and matched property distributions across all data partitions, with mean differences between subsets restricted to less than 5%.

For each agrochemical, three complementary molecular representations were constructed. Molecular graphs were formally represented as $G = (V, E)$, where atoms served as the node set (V), annotated with key atomic-level descriptors, and each explicit chemical bond constituted an undirected edge in the edge set (E) (Table S2). Canonical SMILES strings were encoded into contextual embeddings using ChemFM to capture rich semantic information. Additionally, Molecular ACCESS System (MACCS) fingerprints were computed to encode the presence or absence of key substructural patterns. These representations provided structural, semantic, and functional views of each compound, establishing the multimodal input for the subsequent learning framework.

2.3. Cost-Sensitive Learning

Considering the critical role of data balance in model training, mitigating the detrimental effects of class imbalance was initially explored by processing the training set using the Synthetic Minority Oversampling Technique. However, this technique yielded limited performance benefits (Table S3). Instead, a cost-sensitive learning approach was implemented using a customized weighted focal loss.³⁸ This composite loss formulation integrates two critical mechanisms to optimize model recall for the minority (ecotoxic) class:

2.3.1. Class-Balanced Weight (Pos_Weight). A multiplicative weight was applied to the positive class within the binary cross-entropy loss. This adjustment elevated the penalty for misclassification of minority samples, directly countering the class distribution skew.

2.3.2. Focal Focusing Parameter (γ). A modulating factor was adopted from the focal loss framework: $(1 - p_t)^\gamma$. This factor dynamically scaled the loss to reduce the influence of easily classified negative samples, thereby directing the model's attention toward harder, misclassified examples that may include critical ecotoxic instances.

The combined loss function was formulated on the basis of binary cross-entropy loss with logits, as follows:

$$FL = \alpha \times (1 - p_t)^\gamma \times BCE(\text{inputs, targets})$$

where p_t denotes the estimated probability for the true class, and α acts as a scaling factor. The final loss is computed as the mean over all samples. Consequently, class-level reweighting was integrated with example-level focusing.

2.4. Construction of Multimodal Model and Optimization

The curated data set was split into training, validation, and test sets in an 8:1:1 ratio^{23,24} using the composite stratification variable defined in Section 2.2 (Table S4 and Figure S2). Next, a multimodal deep learning framework was designed to address key challenges in ecotoxicity prediction for bees (i.e., the need for more complete molecular representation and improved generalizability) by comprehensively encoding molecular structure, chemical semantics, and functional attributes (Figure 1). In this architecture, the pretrained chemical LLM served as a frozen feature extractor to leverage prior chemical knowledge, while the GNN and fusion modules were trained end-to-end to optimize the prediction performance. The model concurrently processed molecular graphs, SMILES sequences, and MACCS fingerprints (Table S5) as input modalities. SMILES strings were encoded into rich semantic representations using ChemFM-1B³⁰ (Table S6). The GIN architecture was strategically designated as the graph backbone owing to its theoretical alignment with the Weisfeiler-Lehman isomorphism test and its established empirical strength in capturing complex topological features. Molecular graphs were processed through a 3-layer GIN graph neural network. Batch normalization was also added within the GIN MLP aggregation for stability. Each layer employed a custom multilayer perceptron (MLP) as its internal transformation function, with each MLP incorporating

two successive linear layers to facilitate feature mapping and refinement. The first linear layer projected the concatenated node and semantic features into the hidden dimension, while the second linear layer executed a nonlinear transformation to capture deeper structural dependencies. Both linear layers were integrated with batch normalization and ReLU activation functions.

A progressive fusion mechanism hierarchically integrated the multimodal inputs. Specifically, node-level semantic features derived from ChemFM were iteratively concatenated with atomic representations across successive the graph-encoding stages. Following the final convolution layer, the enriched node-level features were refined through a single node-level attention module that utilized a feedforward network to allocate scalar importance weights to individual atom. These weighted features were aggregated via global mean pooling to produce graph-level embeddings. Subsequently, these structural representations were concatenated with projected fingerprint features to form a unified molecular representation that comprehensively captured complementary chemical information. Stochastic gradient descent was employed during model optimization using a ReduceLROnPlateau scheduler to reduce the learning rate when the validation loss plateaued. Hence, the network was trained to minimize the custom weighted focal loss defined in Section 2.3.

2.5. Comparative Evaluation with Baseline Models

To assess model performance, rigorous benchmarking was conducted against baseline models using the same training, validation, and test splits described in Section 2.4, including tree-based machine learning models (e.g., XGBoost and NGBoost³⁹), standard GNNs (GCN, GIN, GAT), and advanced tabular foundation models, including the recently proposed transformer-based Tabular Prior-data Fitted Network (TabPFN)⁴⁰ and the parameter-efficient ensemble method TabM.⁴¹ Model performance was assessed using a comprehensive set of evaluation metrics, including the area under the receiver operating characteristic curve (AUC), recall, F1-score, specificity, and Matthews correlation coefficient (MCC) (Text S1). In the context of ecotoxicity screening, high recall is crucial for minimizing false negatives, while AUC provides a robust metric of model discrimination across all classification thresholds. Consequently, recall and AUC were prioritized as the primary metrics for model evaluation. All reported results correspond to a single evaluation of the final model, selected based on validation set performance, on the held-out test set. To rigorously evaluate the extrapolative generalizability of the framework, additional benchmarking was conducted using the recently described time split protocol^{35,42} (Figure S8).

2.6. External Validation of Multimodal Prediction Model

To further assess model generalizability, we conducted an additional round of data collection, compiling a comprehensive data set of molecules with documented bee ecotoxicity profiles.^{20,23,24} Following the same precautionary risk-aversion principle adopted during model development, the collected data were curated and labeled in accordance with earlier procedures. This external set was carefully cross-referenced with the modeling data set and duplicate agrochemicals were removed. The remaining 65 unique agrochemicals were employed in an independent external validation.

The external set covered a broad spectrum of agrochemical classes, predominantly comprising herbicides (28%), fungicides (26%), and insecticides (25%), supplemented by other substance types (17%) and plant growth regulators (4%) (Figure S3 and Table S11). This diverse composition reflected real-world use patterns, thereby mitigating potential biases toward any specific agrochemical category during evaluation. Subsequently, predictive performance on this independent collection was assessed to verify the robustness and generalizability of the model to agrochemicals not included in the initial training process.

2.7. Improvement of Model Interpretability

GNExplainer⁴³ was employed to provide transparent and chemically meaningful insights into model predictions by generating atomic- and bond-level attribution maps. The trained GNN was analyzed to identify which atoms and chemical bonds contributed most

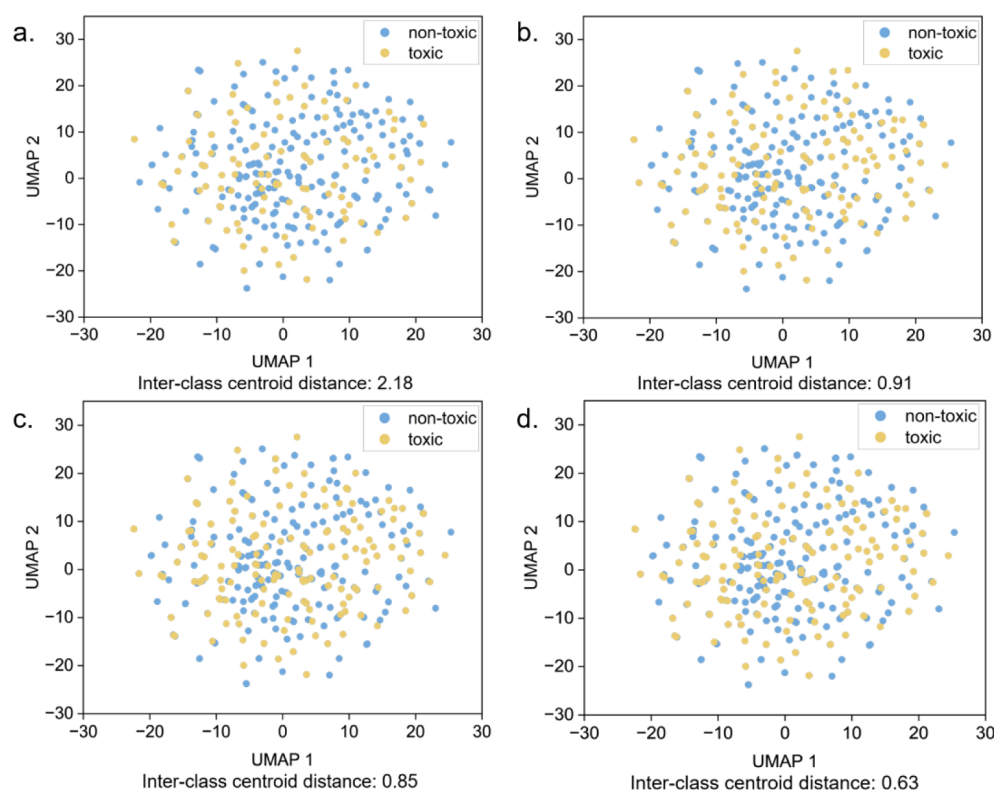


Figure 2. Comparative chemical space analysis across bee ecotoxicity data sets using Uniform Manifold Approximation and Projection (UMAP). High-dimensional Morgan fingerprints are projected into a two-dimensional space for (a) Data set 1, (b) Data set 2, (c) ApisTox data set, and (d) BED. Blue and yellow points represent nontoxic and ecotoxic agrochemicals, respectively. The interclass centroid distance is provided for each data set to quantify the separability between toxicity classes, where a lower value indicates a more integrated and challenging classification landscape.

substantially to each prediction, directly linking model output to the underlying molecular graph topology. In the generated visualizations, atoms were colored on a blue gradient to represent their relative contribution to the prediction, while chemical bonds were highlighted in shades of yellow to indicate their importance for structural recognition. By preserving and explaining the graph structure rather than aggregating features into a global importance score, GNNExplainer delivered instance-specific and structurally grounded explanations.

2.8. Model Deployment on a Cloud Server

The prediction model was deployed as a web-based platform (Figure S4) with a client-server separation pattern to balance usability with computational demands. The front-end interface was constructed using standard web technologies, including HTML, CSS, and JavaScript, which provided an accessible environment for researchers to submit screening queries and visualize results. The back end was implemented within a Python Flask framework, which handled essential operations such as loading the trained prediction model, executing computational workflows, and managing data exchange between components. This organized separation of interface and processing logic contributed to system maintainability and scalable performance for multiuser screening tasks.

2.8.1. Hardware and Software. The Scikit-learn package (v1.6.1) (<https://scikit-learn.org/>) was utilized for data set stratification and conventional model construction. Molecular standardization and feature extraction were performed using RDKit (v2024.3.3) (<https://www.rdkit.org/>). The proposed multimodal framework and baseline graph neural networks were established using PyTorch (v2.6.0) (<https://pytorch.org/>) and PyTorch Geometric (v2.6.1) (<https://pytorch-geometric.readthedocs.io/>), while the extraction of chemical semantic embeddings was facilitated by the Transformers package (v4.53.1) (<https://huggingface.co/>). To support its functionality, all data processing and model training

were performed locally on a workstation computer equipped with an NVIDIA GeForce RTX 4070 Ti SUPER GPU.

3. RESULTS AND DISCUSSION

3.1. Structurally Diverse Chemical Space

Comprehensive data curation yielded the bee ecotoxicity data set (BED), a well-defined and chemically diverse resource for predicting agrochemical ecotoxicity to bees. To evaluate the chemical diversity of BED, its feature distribution was compared against that of three previously published ecotoxicity data sets using Uniform Manifold Approximation and Projection (UMAP) visualization. In Figure 2, each subplot projects high-dimensional molecular fingerprints (ECFP4, 2048 bits) into a two-dimensional space, with blue and yellow points representing nontoxic and ecotoxic agrochemicals, respectively.

Quantitative analysis of the class distribution revealed a notable progression in data complexity across benchmarks. Data set 1²² exhibited an evident boundary between ecotoxicity classes, as evidenced by an interclass centroid distance of 2.18 (Figure 2a). Calculated as the Euclidean distance between the geometric centroids of the ecotoxic and nontoxic classes in the UMAP projection, this metric served as a quantitative proxy for class separability. Data set 2^{23,24} presented an intermediate landscape with an interclass centroid distance of 0.91 (Figure 2b), indicating a moderate degree of mixing between ecotoxic and nontoxic classes. Conversely, the ApisTox data set³⁵ and BED displayed highly integrated distributions. Notably, BED exhibited the tightest

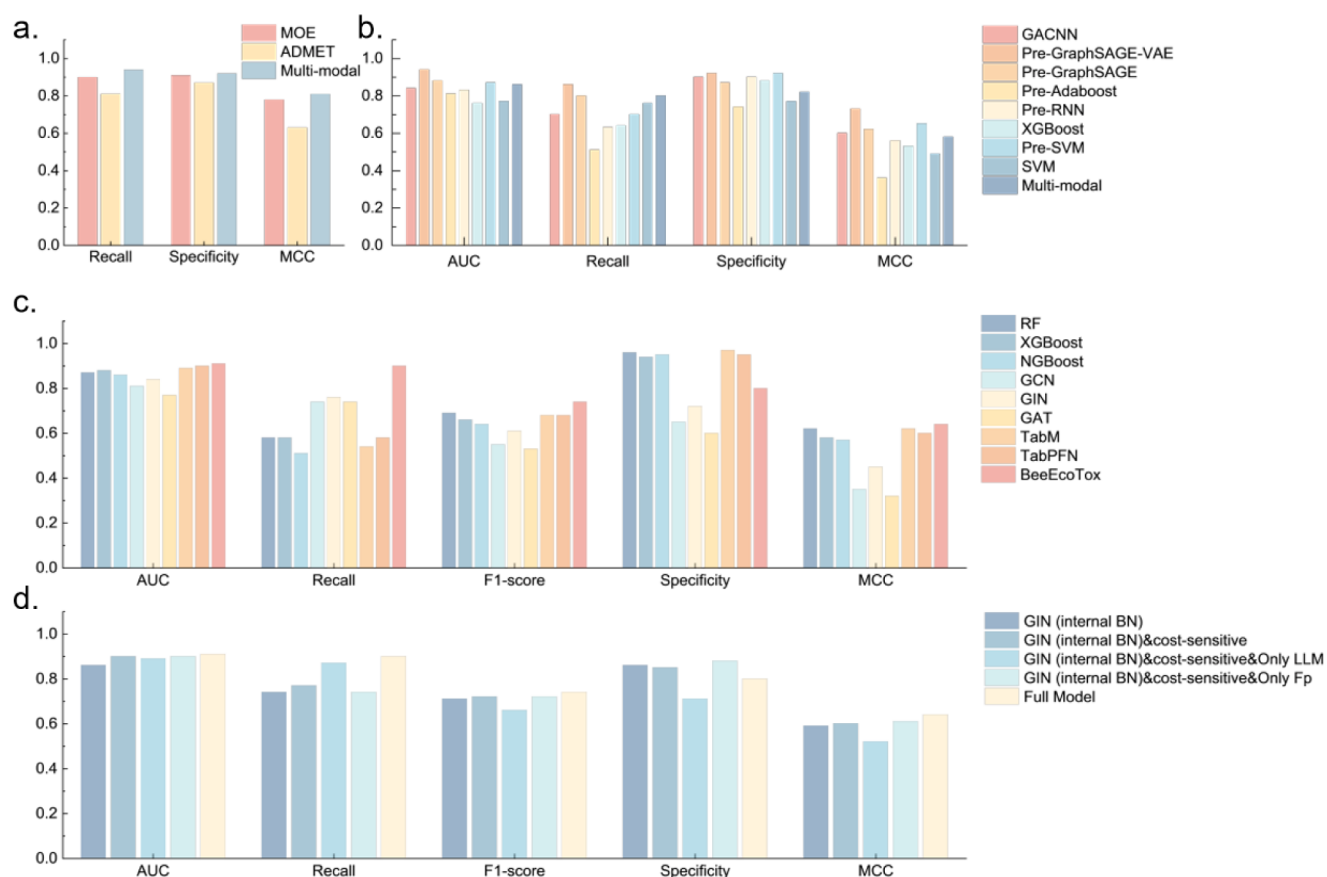


Figure 3. Comprehensive performance evaluation of BeeEcoTox. (a, b) Benchmarking of BeeEcoTox trained on two independent external data sets (Data set 1 and Data set 2), demonstrating its adaptability and competitive performance relative to reported literature baselines. (c) Performance comparison between BeeEcoTox and baseline models on the internal test set, highlighting its state-of-the-art discriminative ability. (d) Ablation study quantifying the contributions of individual components, including structural graph encoding GIN (internal BN), ChemFM-derived semantic embeddings, MACCS fingerprints, and the cost-sensitive learning strategy.

integration, with an interclass centroid distance of 0.63, compared to 0.85 for the ApisTox data set (Figure 2c, d). Results indicated that the ecotoxic and nonecotoxic agrochemicals in BED occupied proximal regions of the feature space, forming a chemically continuous landscape.

Notably, while the ApisTox data set served as a high-quality resource primarily focused on *Apis mellifera*, BED encompassed available ecotoxicity data across diverse bee species under a unified “bees” category. This broader biological classification, combined with the chemical distribution characterized via UMAP analysis, facilitated the development of a predictive model aimed at assessing general agrochemical ecotoxicity to bees.

3.2. Performance of Multimodal Model against Baseline Models

BeeEcoTox integrated molecular graphs through a GIN with internal batch normalization, chemical semantics from SMILES strings encoded by ChemFM, and structural descriptors from MACCS fingerprints. This integrative architecture overcame the limitations of single-modality approaches and exhibited enhanced robustness in data-limited scenarios.

The model achieved state-of-the-art (SOTA) performance (Figure 3c and Table 1), with an AUC value of 0.91, surpassing the performance of standard graph networks by 7%–14% and tree-based machine learning models by 3%–5%.

Table 1. Evaluation Metrics of BeeEcoTox and Baseline Models on the Test Set

	AUC	Recall	F1-score	Specificity	MCC
BeeEcoTox	0.91	0.90	0.74	0.80	0.64
RF	0.87	0.58	0.69	0.96	0.62
XGBoost	0.88	0.58	0.66	0.94	0.58
NGBoost	0.86	0.51	0.64	0.95	0.57
GCN	0.81	0.74	0.55	0.65	0.35
GIN	0.84	0.76	0.61	0.72	0.45
GAT	0.77	0.74	0.53	0.60	0.32
TabM	0.89	0.54	0.68	0.97	0.62
TabPFN	0.90	0.58	0.68	0.95	0.60

High recall is especially critical for ecological risk assessments, as undetected ecotoxicity could lead to continued use of environmentally harmful agrochemicals, posing direct threats to pollinator populations. BeeEcoTox attained a recall value of 0.90, exceeding that of all baseline models by 14%–39%, demonstrating substantially enhanced reliability for prospective hazard identification. This improvement suggested that the multimodal fusion effectively captured subtle toxicophores that single-modality models may overlook. BeeEcoTox achieved an F1-score of 0.74, representing a 5%–21% improvement over comparative methods, confirming robust balance between precision and recall. Notably, while matching the AUC value of advanced tabular foundation models (TabM and TabPFN),

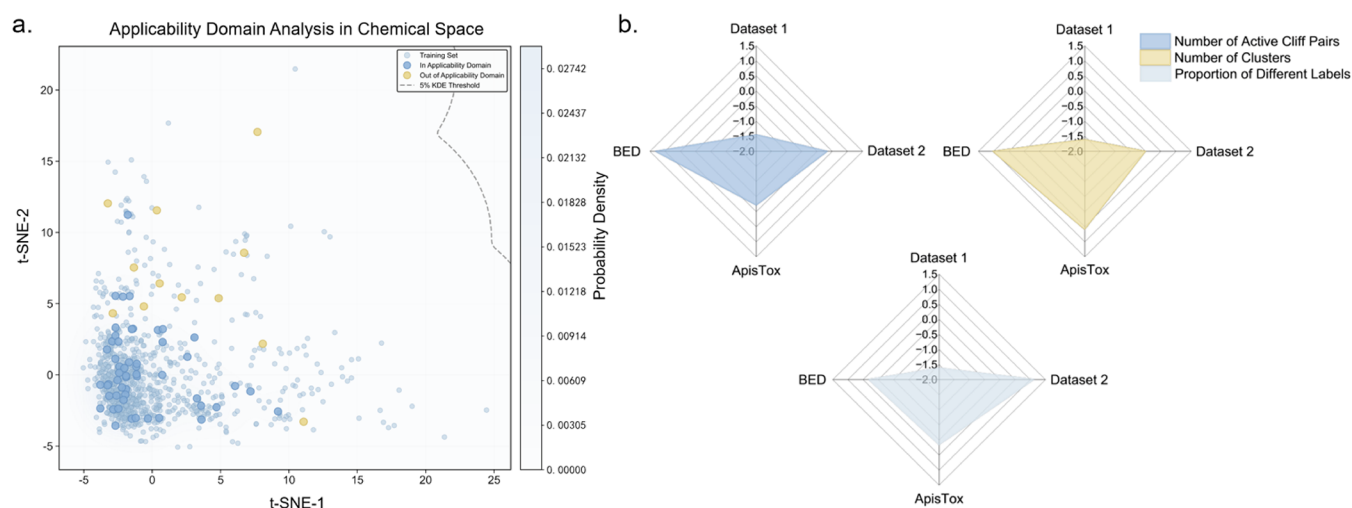


Figure 4. Comparative data landscape analysis and applicability domain (AD) assessment. (a) Kernel density estimation plot characterizing the AD of the framework for external validation. The reliable prediction space is defined by projecting the external validation set onto the training set distribution, with agrochemicals identified as in-domain or out-of-domain based on their probability density within the learned feature space. (b) Ternary plots visualizing the relative complexity of the BED compared to three external benchmarks (Data set 1, Data set 2, and ApisTox), including the number of chemical clusters, the number of activity cliff pairs, and proportion of differently labeled neighbors.

BeeEcoTox achieved 32%–36% higher recall than these architectures, further underscoring its practical advantage in safety screening. Training and validation losses converged stably with only minor fluctuations consistent with expectations for small-scale data sets, and no significant overfitting was observed (Figure S5). As an additional validation step, to verify the statistical robustness of BeeEcoTox, a permutation test ($N = 1000$) was conducted by randomly shuffling training labels and retraining the framework. Results confirmed that BeeEcoTox captures genuine structure-toxicity relationships rather than random correlations, with its original performance significantly exceeding the random distribution ($p < 0.01$) (Figure S6).

The significantly improved performance of BeeEcoTox compared to that of baseline models mainly originated from the synergistic integration of complementary molecular information. Specifically, the baseline GIN model, which incorporates internal batch normalization for stability, established a robust structural foundation with an AUC of 0.86 (Table S9). This structural representation was augmented by ChemFM, which provided contextual, substructure-aware chemical intelligence directly from SMILES strings, while MACCS fingerprints contributed interpretable structural attributes. Combined with cost-sensitive learning, this integrated approach yielded a robust and balanced framework for predicting the ecotoxicity of agrochemicals to bees (Text S2). To further evaluate the adaptability of the proposed framework, an external validation was performed by training the model from scratch on two previously published bee ecotoxicity data sets,^{22–24} using the same data splits and previously reported protocols (Figure 3a, b and Table S7 and S8). BeeEcoTox consistently demonstrated competitive performance, confirming the broad applicability and adaptability of the multimodal fusion strategy across diverse data landscapes and experimental settings.

3.3. Contribution of Multimodal Fusion and Cost-Sensitive Learning

To elucidate the origin of the performance gains detailed above, a systematic ablation study was performed to evaluate

the contribution of each core modality (Figure 3d and Table S9). The baseline model, which solely relied on the GIN with internal batch normalization for structural encoding, established a robust foundation (AUC = 0.86, specificity = 0.86), confirming its capacity to capture fundamental molecular topology. Nevertheless, the recall of the baseline model was constrained (recall = 0.74), revealing a deficit in detecting ecotoxic agrochemicals from the minority class. The incorporation of the cost-sensitive learning approach effectively mitigated class imbalance, elevating the AUC and recall values to 0.90 and 0.77, respectively while maintaining robust specificity. This step underscored the effectiveness of loss reweighting in redirecting the attention of the model toward ecotoxic instances without compromising overall discrimination.

To further justify the specific design choice of internal batch normalization within the GIN architecture, a separate, dedicated ablation study was executed (Table S10). This analysis, performed under identical cost-sensitive learning conditions, directly compared a pure GIN, a GIN with batch normalization between convolutional layers, and the proposed GIN with batch normalization inside the MLP. The results revealed that the pure GIN suffered from severe mode collapse due to data imbalance. Conversely, integrating batch normalization inside the MLP provided the most stable and superior discriminative performance compared to the conventional placement between layers.

To mitigate the performance trade-off introduced by loss reweighting alone, complementary molecular representations were introduced. The addition of semantic embeddings derived from ChemFM proved particularly decisive. These embeddings capture implicit, context-aware chemical features directly from SMILES strings, providing complementary representations for explicit topological features.^{44,45} Incorporating these semantic features shifted the focus of the model toward hazard identification, drastically elevating the recall value to 0.87. This finding demonstrated that chemical language semantics were highly effective at flagging subtle toxicophores, although this heightened recall resulted in a

specificity trade-off (specificity = 0.71). In comparison, the inclusion of MACCS fingerprints prioritized discriminatory precision. The model variant utilizing these explicit descriptors achieved a robust specificity of 0.88, indicating that predefined substructural patterns acted as a conservative filter to minimize false positives by enforcing strict structural constraints.

The full integration of structural, semantic, and fingerprint representations, together with the cost-sensitive learning approach, yielded a balanced and discriminative model. This comprehensive approach demonstrated that predicting bee ecotoxicity substantially benefited from a multimodal learning strategy. Molecular graphs defined the topological framework of agrochemicals, while semantic embeddings contributed contextual chemical intelligence to their substructures. Combined with MACCS fingerprints that encoded discrete functional group information, these complementary representations yielded a more integrative molecular profile. The resulting multimodal characterization better resolved complex and subtle structure-toxicity relationships. The progressive performance improvements observed throughout the ablation study highlighted the critical importance of combining complementary feature types while addressing class imbalance, building robust predictors for bee ecotoxicity through the effective integration of multiple representational modalities.

3.4. Model Generalizability via External Validation

To rigorously evaluate the generalizability of BeeEcoTox, an independent external validation was conducted using a set of agrochemicals excluded from model development. The predictive reliability of the model was assessed using a kernel density estimation (KDE)-based applicability domain (AD) framework.³⁷ Each agrochemical was projected into a high-dimensional feature space learned by the model and its probability density was estimated relative to the training distribution, with the in-domain threshold defined as the fifth percentile density of the training set (Figure 4a). This domain assessment strategy provided critical interpretability for practical risk-screening applications.

On the entire external set, BeeEcoTox achieved AUC and accuracy values of 0.85 and 0.83, respectively, demonstrating robust overall discriminative ability between ecotoxic and nonecotoxic agrochemicals (Table S11). A detailed AD assessment revealed that 81.5% of agrochemicals (53 of 65) resided within the well-characterized chemical space. Model performance was robust and balanced for these in-domain agrochemicals, with AUC, recall, and accuracy values of 0.87, 0.82, and 0.85, respectively. By contrast, the remaining 18.5% of agrochemicals (12 of 65) were flagged as out-of-domain, indicating higher structural novelty relative to the training corpus. While the model yielded correct predictions for most cases, the limited sample size precluded reaching definitive statistical conclusions regarding its extrapolation performance. The variability in performance underscored the expected increase in predictive uncertainty when operating outside the core AD. Importantly, the consistently high performance for the majority of in-domain agrochemicals confirmed the reliability of the model for screening structurally related agrochemicals. Simultaneously, explicit identification of out-of-domain agrochemicals alerted users to perform additional verifications, offering a transparent and precautionary mechanism.

3.5. Efficient Identification and Alerts of Structural Toxicophores via GNNExplainer

GNNExplainer was employed to derive atomic attribution maps and identify critical toxicophores. Although this methodology inherently confines interpretability to topological representations, a global weight assessment corroborated the predominant influence of the graph branch relative to molecular fingerprints. This validation ensured that the visualized motifs truthfully reflected the primary determinants of model predictions (Figure S7). The model revealed key structural motifs that drive bee toxicity predictions across major insecticide classes, including organophosphate, carbamate, neonicotinoid, and pyrethroid insecticides, aligning with reportedly pronounced ecotoxicity risks to bees compared to herbicides and fungicides.^{22,46} As illustrated in Figure 5,

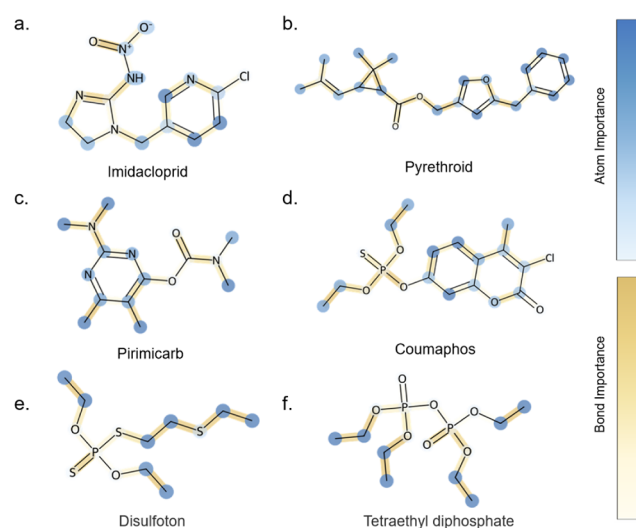


Figure 5. GNNExplainer-based identification of ecotoxicity-determining substructures in representative insecticides. Atom- and bond-level attribution maps highlight structural motifs that the model associates with bee ecotoxicity. (a) Imidacloprid (neonicotinoid insecticide). (b) Pyrethroid insecticide containing aromatic/heterocyclic rings. (c) Pirimicarb (carbamate insecticide). (d) Coumaphos (organophosphate insecticide). (e) Disulfoton (organophosphate insecticide). (f) Tetraethyl diphosphate (organophosphate insecticide).

organophosphate insecticides were identified to exert ecotoxic effects on bees primarily through the irreversible inhibition of acetylcholinesterase (AChE), leading to the accumulation of acetylcholine and subsequent neuroexcitation. Motifs containing P–S or P=S bonds were consistently linked to enhanced pesticidal activity and bee ecotoxicity by the model (Figure 5d–f). These phosphorothioate groups often act as pro-toxicants, requiring *in vivo* oxidative desulfuration to the corresponding phosphate (P=O) for potent AChE inhibition, a key activation step in the mode of action of organophosphate insecticides.^{47–50} Additionally, the substitution of alkyl core scaffolds with aromatic rings was predicted to increase their ecotoxicity, consistent with reports that reduced molecular flexibility and increased planarity improved target specificity and enhanced ecotoxicity to bees.^{47,48}

Among neonicotinoid insecticides, three functional groups (nitroguanidine, cyanoamidine, and butenolide) were consistently identified as ecotoxicity-determining motifs, aligning with reported adverse effects in nontarget organisms.^{47,51} In alignment with this observation, imidacloprid (Figure 5a),

which contains a nitroguanidine pharmacophore, has been confirmed to exhibit significantly higher toxicity to bees compared to other subgroups.⁴⁷ Additionally, the model associated pyrethroid insecticides, which contain aromatic or heterocyclic rings (Figure 5b), with bee ecotoxicity, aligning with previous studies that correlated these motifs with adverse effects on bee health.^{23,47,52}

Notably, the carbamate insecticide pirimicarb (Figure 5c) was confirmed to exhibit oral acute toxicity to bees, with potential implications for critical behavioral functions,⁵³ supporting earlier risk assessments conducted in oilseed rape environments that highlighted its role as a phenyl derivative capable of inducing notable toxicity under relevant field exposure scenarios.^{47,54}

3.6. Data Landscape and Modeling Challenges: Activity Cliffs and Structural Diversity

The predictive efficacy of ecotoxicity models is intrinsically linked to the chemical and toxicological landscapes of the training data. Consequently, key data properties were evaluated to systematically assess this relationship, including structural diversity and the prevalence of activity cliffs,^{55,56} which are defined as pairs of structurally similar compounds that exhibit marked differences in ecotoxicity. As summarized in the ternary plots in Figure 4b (Text S3), BED exhibited greater structural diversity and a higher density of activity cliffs compared to earlier benchmark data sets, reflecting a more realistic and chemically complex space that closely mirrored real-world agrochemical scenarios.

This elevated complexity presents a greater learning challenge for models to distinguish subtle structural features associated with divergent ecotoxicity outcomes. Notably, BeeEcoTox achieved competitive performance within this challenging environment, demonstrating that its multimodal fusion strategy provided a robust framework for navigating intricate structure–activity landscapes. The observed correlation between increasing data set complexity and performance decline across benchmark comparisons (Tables S7 and S8) underscored the general challenge in this field.

A recent benchmark study utilizing the ApisTox data set highlighted the importance of rigorous temporal splitting for evaluating extrapolative generalizability.⁴² To investigate this phenomenon, the reported time split protocol was applied to BED. Results revealed a notable decline in predictive performance, specifically, the AUC value decreased from 0.91 to 0.72 and the MCC value dropped to 0.22. This declining trend paralleled the performance decay observed in high-capacity models in the benchmark study, which was attributed to severe distribution shifts characterized by a 3-fold increase in chemical space distance (Figure S8 and Table S12). Although these distributional discrepancies currently impede overall discrimination in data-limited scenarios, such rigorous protocols offer valuable real-world relevance and will likely serve as a definitive standard for prospective validation as ecotoxicological data become increasingly available.

3.7. Application of BeeEcoTox through a Cloud-Based Platform

BeeEcoTox was deployed as a publicly available web-based platform (<https://www.ai4environ.cn/BeeEcoTox>) designed to support chemical safety assessments in agricultural contexts. The platform supports single-compound queries and batch processing, enabling rapid screening of agrochemicals and environmentally relevant compounds. The interface presents

binary ecotoxicity classifications with intuitive color coding, using red and green to indicate ecotoxic and nonecotoxic agrochemicals to bees, respectively. The platform provides substructure-level visualization, highlighting atoms and bonds identified as critical structural determinants for bee ecotoxicity, alongside AD analysis, offering users mechanistic insights and reliability estimates for each prediction while illustrating model interpretability. This platform can assist researchers and regulators in making evidence-based decisions on agrochemical safety and promote the development of new approach methodologies by enabling efficient *in silico* prioritization for subsequent experimental validation, thus supporting sustainable agricultural practices and pollinator protection strategies.

4. ENVIRONMENTAL IMPLICATION

BeeEcoTox is a multimodal deep learning model for evaluating agrochemical ecotoxicity to bees in a high-throughput mode. The semantically enriched graph-learning approach integrates SMILES-derived embeddings from a pretrained chemical LLM, ChemFM, into a GIN with internal batch normalization. The architecture fuses structural, semantic, and fingerprint-based molecular representations, achieving SOTA performance against a suite of baseline models for interpretable and robust prediction of agrochemical ecotoxicity to bees. Future efforts to enhance ecological realism and regulatory applicability should focus on refining exposure pathway specificity between contact and oral uptake, integrating quantitative exposure metrics, and broadening the chemical coverage and annotation consistency of bee ecotoxicity data sets. Overall, BeeEcoTox serves as an effective screening tool for ecological risk assessments of agrochemicals, providing evidence-based support for reliable regulation.

■ ASSOCIATED CONTENT

Data Availability Statement

The code for model construction is available from the corresponding author on reasonable request. Toxicity data from the Pesticide Properties Database (PPDB) can be accessed directly at <https://sitem.herts.ac.uk/aeru/ppdb/en/>.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.5c18162>.

Detailed descriptions of model evaluation performance metrics (Text S1); machine learning algorithms and hyperparameter optimization (Text S2); chemical landscape and activity cliff analyses (Text S3); data set information (Table S1); initial atom and bond features for graph representation (Table S2); performance comparisons with and without oversampling (Table S3); data partition results after stratification (Table S4); comparison of molecular fingerprint representations (Table S5); evaluations of various chemical language model encoders (Table S6); comparative assessments against literature benchmarks (Tables S7 and S8); ablation study of framework components (Table S9); ablation study on the placement of Batch Normalization in GIN (Table S10); external validation results (Table S11); comparative assessment of model performance under property-based and temporal splitting protocols (Table S12) (XLSX)

Venn diagram illustrating the overlap and composition between the bee ecotoxicity data set (BED) curated in

this study and the ApisTox benchmark (Figure S1); visualizations illustrating the data processing pipeline (Figure S2); class distribution of the external validation set (Figure S3); architecture of the web platform (Figure S4); training loss curve (Figure S5); statistical validation via permutation testing (Figure S6); global weight-based interpretability analysis (Figure S7); quantitative analysis of data splitting strategies (Figure S8) (PDF)

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Notes

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